Notes on numerical solution of OF-DFT Equation

Orbital-Free density equation

$$\left[-\frac{1}{2}\nabla^2 + V_\theta(\vec{r}) + V_{KS}(\vec{r})\right]\phi(r) = \mu\,\phi(r)$$

where density $n(r) = \phi^2(r)$

$$V_\theta = \frac{\delta T_\theta}{\delta n}$$

$$V_{KS} = V_{ext} + V_H + V_{xc}$$

$$V_{ext} = \frac{-Z}{r}$$

$$V_H = \int \frac{n(\vec{r}')}{|r-\vec{r}'|}\,d\vec{r}'$$

$$V_{xc} = \frac{\delta E_{xc}}{\delta \rho}$$

We will use approximate forms for V_θ and V_{xc} in t
of density $n(\vec{r})$

We consider closed atoms (He, Ne, Ar, Xe, Kr)
for closed shell atoms

$$\phi(\vec{r}) = R(r)\,Y_{lm}(\theta,\phi)$$

for ground state $l = 0$ $m = 0$

$$\phi(\vec{r}) = \phi(r)$$

using representation for ∇^2 in spherical coordinate syste

$$\nabla^2 = \frac{1}{2}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) = \frac{1}{r^2}\left[2r\frac{\partial}{\partial r} + r^2\frac{\partial^2}{\partial r^2}\right]$$

$$= \frac{d^2}{dr^2} + \frac{2}{r}\frac{d}{dr}$$

we write

$$\left[\frac{-1}{2}\left(\frac{d^2\phi}{dr^2} + \frac{2}{r}\frac{d\phi}{dr}\right) + v_0(r) + V_{ext}(\vec{r}) + v_{xc}(r) + v_H(r)\right]\phi(r)$$

$$\mu\,\phi(r)$$

first we change the variable from r to x given by

$$r = x^2$$

$\because \frac{dr}{dx} = 2x$

$\frac{dx}{dr} = \frac{1}{2x}$

$\boxed{\frac{d}{dr} = \frac{d}{dx}\cdot\frac{1}{2x}}$

This transformation leads to

$$\frac{d}{dr} = \frac{1}{2x}\frac{d}{dx}$$

and

$$\frac{d^2}{dr^2} = \frac{1}{4x^2}\frac{d^2}{dx^2} - \frac{1}{4x^3}\frac{d}{dx}$$

$$\therefore \frac{d^2\phi}{dr^2} + \frac{2}{r}\frac{d\phi}{dr} = \frac{1}{4x^2}\frac{d^2\phi}{dx^2} - \frac{1}{4x^3}\frac{d\phi}{dx} + \frac{2}{x^2}\times\frac{1}{2x}\frac{d\phi}{dx}$$

$$= \frac{1}{4x^2}\frac{d^2\phi}{dx^2} + \frac{3}{4x^3}\frac{d\phi}{dx}$$

$\therefore$ OF equation becomes

$$-\frac{1}{2}\left(\frac{1}{4x^2}\frac{d^2\phi}{dx^2} + \frac{3}{4x^3}\frac{d\phi}{dx}\right) + V_\theta\,\phi(x) + V_{KS}\,\phi(x) = \mu\phi(x)$$

$$-\frac{1}{8x^2}\frac{d^2\phi}{dx^2} - \frac{3}{8x^3}\frac{d\phi}{dx} + V_\theta(x)\,\phi(x) + V_{KS}(x)\,\phi(x) = \mu\phi(x)$$

$$\frac{d^2\phi}{dx^2} + \frac{3}{8x^3}\times 8x^2\frac{d\phi}{dx} - 8x^2\left(V_\theta(x) + V_{KS}(x)\right)\phi(x) = (8x^2)\,\mu\phi(x)$$

$$\frac{d^2\phi}{dx^2} + \frac{3}{x}\frac{d\phi}{dx} - \left(V_\theta(x)\,8x^2 + V_{KS}\,8x^2\right)\phi(x) + \mu\,8x^2\phi(x) = 0$$

$$\boxed{\frac{d^2\phi}{dx^2} + \frac{3}{x}\frac{d\phi}{dx} + \left(\mu - V_\theta(x) - V_{KS}(x)\right)8x^2\,\phi(x) = 0}$$

This equation is in the form

$$\frac{d^2\phi}{dx^2} + P(x)\frac{d\phi}{dx} + Q(x)\,\phi(x) = 0$$

where

$$P(x) = \frac{3}{x}$$

$$Q(x) = (\mu - V_0(x) - v_{KS}(x))\, 8x^2$$

We cast it in the form

$$y'' + q(x)\, y(x) = 0$$

where

$$\phi(x) = y(x) f(x)$$

with $f = \frac{1}{x^{3/2}}$

and

$$q(x) = Q(x) - \frac{P^2(x)}{4} - \frac{P'(x)}{2}$$

$$q(x) = (\mu - V_0(x) - v_{KS}(x))\, 8x^2 - \frac{1}{4}\left(\frac{9}{x^2}\right) - \frac{1}{2}\left(\frac{-3}{x^2}\right)$$

$$= (\mu - V_0(x) - v_{KS}(x))\, 8x^2 - \frac{9}{4x^2} + \frac{1 \times 3}{2x^2}$$

$$q(x) = (\mu - V_0(x) - v_{KS}(x))\, 8x^2 - \frac{3}{4x^2}$$

$$\therefore \frac{d^2y}{dx^2} + \left((\mu - V_0(x) - V_{KS}(x))\, 8x^2 - \frac{3}{4x^2}\right) y = 0$$

$$= \frac{d^2y}{dx^2} + \left(\mu - V_0(x) - V_{KS}(x) - \frac{3}{4x^2} \times \frac{1}{8x^2}\right) 8x^2 y = 0$$

$$\frac{d^2y}{dx^2} + \left(\mu - V_0(x) - V_{KS}(x) - \frac{3}{32x^4}\right) 8x^2 y = 0$$

$$\boxed{\frac{d^2y}{dx^2} + \left(\mu - V_{eff}(x)\right) y\, 8x^2 = 0}$$

$\langle \psi | \hat{H} | \psi \rangle$

$\parallel$

where

$$V_{eff} = \left(V_0(x) + V_{KS}(x) + \frac{3}{32x^4}\right)$$

and

$$\phi(x) = \frac{y(x)}{x^{3/2}} = \frac{y(x)}{x^2}$$

Normalization condition

$$\int n(\vec{r})\, d^3\vec{r} = 4\pi \int n(r)\, r^2 dr = N$$

$$4\pi \int \phi^2(r)\, r^2 dr = 4\pi \int \phi^2(x)\, x^4 (2x\, dx)$$

$$= 4\pi \int \phi^2(x)\, 2x^5 dx$$

$$= 4\pi \int 2\phi^2(x)\, x^5 dx$$

$$= 8\pi \int \frac{y^2(x)}{x^3}\, x^5 dx$$

$$\int n(\vec{r})\, d^3\vec{r} = 8\pi \int y^2(x)\, x^2\, dx = N$$

Expression for the potential

within Thomas - Fermi approximation

$$v_0 = C_k \frac{5}{3} n^{2/3} \qquad C_k = \frac{3}{10}(3\pi^2)^{2/3}$$

$$v_{KS} = v_{ext} + v_H + v_x + v_c$$

within LDA

$$v_x = -C_x \frac{4}{3} n^{1/3} \qquad C_x = \frac{3}{4}\left(\frac{3}{\pi}\right)^{1/3}$$

$$v_{ext} = -\frac{Z}{x^2}$$

$$v_H = \int \frac{n(\vec{r}')}{|\vec{r}-r'|}\, d\vec{r}'$$

Gunnarsson - Lundqvist (GL) correlation potential

$$v_c = -0.0666 \ln\left(1 + \frac{11.4}{r_s(\vec{r})}\right)$$

$$r_s(\vec{r}) = \left(\frac{3}{4\pi}\frac{1}{n(r)}\right)^{1/3}$$

Expression for V_H

$$V_H = \int \frac{n(\vec{r}')}{|r-\vec{r}'|} d\vec{r}'$$

using

$$\frac{1}{|\vec{r}-\vec{r}'|} = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{4\pi}{(2l+1)} \frac{(r_<)^l}{(r_>)^{l+1}} Y^*_{lm}(\theta,\phi) Y_{lm}(\theta',\phi')$$

where $r_<$ is the smaller of r and r'
and $r_>$ is the larger of r and r'

using the fact $n(\vec{r}') = n(r')$ (independent of θ and ϕ)

$$V_H = \int n(r') \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{4\pi}{(2l+1)} \frac{(r_<)^l}{(r_>)^{l+1}} Y^*_{lm}(\theta,\phi) Y_{lm}(\theta'\phi') r'^2 \cdot dr' d\Omega'$$

$$= \cancel{4\pi \int n(r') \frac{r_<^0}{r_>^{0+1}} r'^2 dr' Y_{00}(\theta\phi)\sqrt{4\pi}}$$

$$= \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{4\pi}{(2l+1)} \int n(r') \frac{r_<^l}{r_>^{l+1}} r'^2 dr' \int d\Omega' Y_{lm}(\theta\phi) Y_{lm}(\theta'\phi') \sqrt{4\pi}\, Y_{00}(\theta'\phi') d\Omega$$

$$= 4\pi \int n(r') \frac{r_<^0}{r_>^{0+1}} r'^2 dr' Y_{00}(\theta\phi) \sqrt{4\pi} \qquad \left(Y_{00}(\theta\phi) = \frac{1}{\sqrt{4\pi}}\right)$$

$$= 4\pi \int_0^{\infty} n(r') \frac{1}{r} r'^2 dr' = 4\pi \int_0^{r} n(r') \frac{1}{r} r'^2 dr'$$

$$+ 4\pi \int_r^{\infty} n(r') \frac{1}{r'} r'^2 dr$$

$$V_H = 4\pi \left[\frac{1}{r} \int_0^{r} n(r') r'^2 dr' + \int_r^{\infty} n(r') r' dr' \right]$$

from normalization condⁿ

$$\int n(\vec{r}') d^3\vec{r}' = N$$

$$4\pi \int_0^{\infty} n(r') r'^2 dr' = N$$

$$4\pi \int_0^{r} n(r') r'^2 dr' + 4\pi \int_r^{\infty} n(r') r'^2 dr' = N$$

$$\therefore 4\pi \int_0^{r} n(r') r'^2 dr' = N - 4\pi \int_r^{\infty} n(r') r'^2 dr'$$

$$\therefore V_H = 4\pi \left[-\frac{1}{r} \int_r^{\infty} n(r') r'^2 dr' \right] + \frac{N}{r} + 4\pi \int_r^{\infty} n(r') r' d$$

$$V_H = \frac{N}{r} + 4\pi \left[\int_r^{\infty} n(r') r' dr' - \frac{1}{r} \int_r^{\infty} n(r') r'^2 dr' \right]$$

using $n(r') = \phi^2(r')$

and $r' = x'^2 \qquad r = x^2$

$$V_H = \frac{N}{x^2} + 4\pi\left[\int_x^\infty n(x')\,x'^2\,2x'\,dx' - \frac{1}{x^2}\int_x^\infty n(x')\,x'^4\,2x'\,dx'\right]$$

$$V_H = \frac{N}{x^2} + 8\pi\left[\int_x^\infty n(x')\,x'^3\,dx' - \frac{1}{x^2}\int_x^\infty n(x')\,x'^5\,dx'\right]$$